

9: A model Rocket

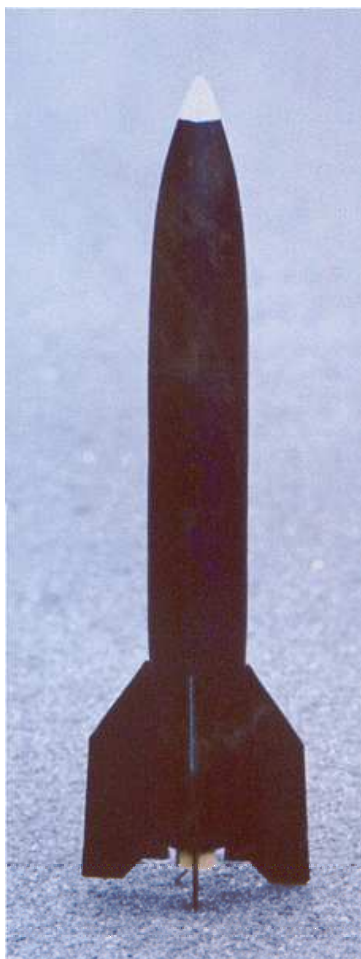


Figure 9.1: A model of the German rocket V2 from world war 2. The length is 40 cm and the diameter 4.8 cm. It is equipped with an ESTES B4-2 engine. This should deliver a modest thrust of 4 Newton in 1.2 sec

The rocket seen in fig. 9.1 is a model of the German V2 from world war 2, build by one of my students, Jonas. Using the merry-go-round Jonas found that it has a drag coefficient of $C_d = 0.23$. Then he used the merry-go-round in a way which has not been mentioned earlier in these notes:

He fired the engine with the rocket fixed to the merry-go-round. Via a large 72-tooth cock wheel (see fig. 9.2) and the photo gate from Pascoe's Smart Pulley, he was then able to determine position $s(t)$, velocity $v(t)$ and acceleration $a(t)$ for the movement along the circular s-axis. Knowing the mass as well as the acceleration of the equivalent car,

the thrust of the engine could be determined as a function of time.

On the background of the data obtained, it became possible to simulate the flight of the rocket on computer. At the time we where not able to check with measurements of the real launch. At moment I do not have the time to present and analyse the data he obtained. I can however recommend the interested reader to have a look at Jonas' fine home page www.serg.dk, to see how far his continued interest in model rockets has brought him.

Those with an interest in the possibility of model rocketry at schools will appreciate S. A. Widmark's article *Rocket Physics* in *The Physics Teacher* (March 1998). Much more stuff on the subject can be found on NASA's home page¹. In one point, my measurements disagree with Widmarks: By use of a primitive wind tunnel - a wet/dry vacuum cleaner sucking air through a pipe - he finds values of C_d as being about 1.0 - 1.5. This is much more than the 0.23 found for the rocket in fig. 9.1. My guess is that Widmarks pipe is too narrow. However, other high values can be found on other home pages as being used to explain the measured altitudes in actual launches. I have not yet found the reason for these discrepancies.

¹ www.grc.nasa.gov/WWW/K-12/airplane/dragco.html



Figure 9.2: From this angle one can see the 72-tooth cock wheel underneath the merry-go-round. To the left one sees the metal brace from fig. 5.3, used for mounting the merry-go-round with a horizontal axis. Other details from fig. 5.3 are recognised too. Furthermore one sees the two horizontal springs from fig. 5.2a, suspended from a horizontal arm, ready for determination of the period for the harmonic oscillation. The same springs can be moved to another vertical arm, hidden by the brace, ready for the measurement described in fig. 5.2a.